§14.2 Limits and Continuity

Homework： 7,12,13,17,19,22,23,29,32,35,37,39

(1) 極限： $f: R^2 \to R$. Its domain $= D$

$$\lim_{(x,y)\to(a,b)} f(x,y) = L$$

Definition：Given (any) $\varepsilon > 0$, $\exists$ a $\delta > 0$ s.t.

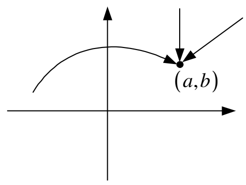
$$|f(x,y) - L| < \varepsilon \text{ whenever } 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$$

and $(x,y) \in D$.

(i) 極限存在必唯一

(ii) $(x,y) \to (a,b)$ (二維逼近)和 $x \to a$ (一維逼近)的差別

- 在 2 維有無窮多的方向逼近到 (a,b)
- 沿著曲線也可以



(iii)驗證不存在：只需找 2 個方向，其極限值不同。

(iv)驗證存在：定義或夾擊或 polar coordinate.

Example 1： $f(x,y) = \dfrac{x^2 - y^2}{x^2 + y^2}$. Show that $\lim\limits_{(x,y)\to(0,0)}$ does not exist.

Proof：

Key：分子和分母為齊次(各項次數相同)，則可沿直線証明極限不存在。

Let： $y = mx \Rightarrow f(x,mx) = \dfrac{1-m^2}{1+m^2}$.

$$\lim_{(x,mx)\to(0,0)} f(x,mx) = \lim_{(x,mx)\to(0,0)} \frac{1-m^2}{1+m^2} = \lim_{x\to 0} \frac{1-m^2}{1+m^2} = \frac{1-m^2}{1+m^2}$$

隨 m 而變的極限 $\Rightarrow$ 不存在

Example 2： Show that $\lim\limits_{(x,y)\to(0,0)} \dfrac{xy^2}{x^2 + y^4}$ does not exist.

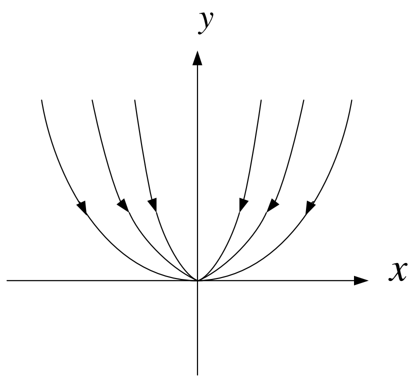
Proof：

Key：分子和分母各項如何變齊次。

Let： $x = my^2 \Rightarrow \dfrac{xy^2}{x^2+y^4} = \dfrac{my^4}{m^2y^4+y^4} = \dfrac{m}{1+m^2}$

$\Rightarrow$沿著 $x = my^2$ 的拋物線趨近 $(0,0)$ 則其極限為 $\dfrac{m}{1+m^2}$ （隨 m 而變）

$\Rightarrow$不存在



Example 3： Prove that $\displaystyle\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0$

Proof：

$$0 \le \left|\frac{3x^2y}{x^2+y^2}\right| \le 3|y|$$

當 $(x,y)\to(0,0)$，上、下界的極限皆趨近於 0

$$\Rightarrow \lim_{(x,y)\to(0,0)} \left|\frac{3x^2y}{x^2+y^2}\right| = 0$$

$$\Rightarrow \lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = 0$$

Example 4： Prove that $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2y^3}{x^2+y^2} = 0$

(1)By definition：Skip.

(2)Sequence Theorem：

$$0 \le \left|\frac{x^2y^3}{x^2+y^2}\right| \le \left|y^3\right|$$

$$\begin{array}{ccc} \swarrow & \Downarrow & \searrow \\ 0 & & 0 \end{array}$$

$$\longrightarrow 0$$

(3)Polar coordinate $x = r\cos\theta$, $y = r\sin\theta$.

$$\frac{x^2y^3}{x^2+y^2} = r^3\cos^3\theta\sin^3\theta \to 0 \text{ as } (x,y)\to 0 \text{ (and so } x^2+y^2=r\to 0\text{)}$$

Example 5： Decide whether $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^3+y^2}{x^2+y^2}$ exist or not by polar coordinate.

Solution：

$$\frac{x^3+y^2}{x^2+y^2} = \frac{r^3\cos^3\theta + r^2\sin^2\theta}{r^2} = r\cos^3\theta + \sin^2\theta$$

$$\to \sin^2\theta \text{ as } r\to 0 \Rightarrow \text{不存在}$$

Example 6： Decide whether $\lim\limits_{(x,y)\to(0,0)} \dfrac{x^2+\sin^2 y}{2x^2+y^2}$ exist or not.

Solution：

Let $y = mx$

$$\Rightarrow \frac{x^2+\sin^2 y}{2x^2+y^2} = \frac{x^2 + \dfrac{\sin^2 mx}{m^2x^2}m^2x^2}{2x^2+m^2x^2} = \frac{1+\left(\dfrac{\sin mx}{mx}\right)^2 m^2}{2+m^2}$$

$$\to \frac{1+m^2}{2+m^2} \text{ as } x\to 0 \Rightarrow \text{不存在}$$

(此例子可當成齊次式)

Summary：

$\lim\limits_{(x,y)\to(0,0)} \dfrac{f}{g}$, where f and g are polynomials in x and y.

(1) g 的某一項的次數 $\geq f$ 的某一項的次數. $\Rightarrow$ 不存在.

(2) $g(x,y)\neq 0$ for all $(x,y)\neq(0,0)$ 和 g 的每一項的次數 $< f$ 的每一項次數.
$\Rightarrow$ 存在.

Example 7： Decide whether $\lim\limits_{(x,y)\to(0,0)} \dfrac{x+y^3}{x^2+y^2}$ exist or not by polar coordinate.

Solution：

$$\lim_{(x,y)\to(0,0)} \frac{x+y^3}{x^2+y^2}$$

$$= \lim_{r\to 0} \frac{r\cos\theta + r^3\sin^3\theta}{r^2}$$

$$= \lim_{r\to 0}\left(\frac{\cos\theta}{r} + r\sin^3\theta\right) \Rightarrow \text{不存在} .$$

Example 8： Decide whether $\lim_{(x,y)\to(0,0)} \dfrac{x^2+y^2}{x+y^3}$ exist or not by polar coordinate.

Solution：

$$\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{x+y^3}$$

$$=\lim_{r\to 0} \frac{r}{\cos\theta + r^2\sin^3\theta}$$

$$=\lim_{r\to 0}\left(\frac{1}{\dfrac{\cos\theta}{r}+r\sin^3\theta}\right)=\begin{cases}\text{不存在}. & \theta=\dfrac{\pi}{2}\\ 0 & \theta\neq\dfrac{\pi}{2}\end{cases} \Rightarrow \text{不存在}.$$

Example 9： Prove that $\lim_{(x,y,z)\to(0,0,0)} \dfrac{xyz}{x^2+y^2+z^2}=0$

Proof：

$$\frac{x^2+y^2+z^2}{3}\geq\sqrt[3]{x^2y^2z^2}=\left(xyz\right)^{\frac{2}{3}}$$

$$\Rightarrow 0\leq\left|\frac{xyz}{x^2+y^2+z^2}\right|\leq\left|\frac{xyz}{3\left(xyz\right)^{\frac{2}{3}}}\right|=\frac{1}{3}\bullet\left|xyz\right|^{\frac{1}{3}} \to 0$$

as $\left(x,y,z\right)\to\left(0,0,0\right)$.

(2) f is continuous at $\left(a,b\right)$ provided that $\lim_{(x,y)\to(a,b)} f\left(x,y\right)=f\left(a,b\right)$.

Example 10： Let $f\left(x,y\right)=x^2+y^2$. f is continuous at $\left(0,0\right)$.

Example 11： $f\left(x,y\right)=\ln\dfrac{y}{x}$

Domain：$\left\{\ \left(x,y\right):xy>0\ \right\}$

f is continuous on its domain.

Example 12：

$$f(x,y)=\begin{cases} \dfrac{x^2+y^2}{\sqrt{x^2+y^2+1}\ -1} & (x,y)\neq(0,0) \\ 0 & (x,y)=(0,0) \end{cases}$$

Find the set of points of continuity of f .

Solution：

$$\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+1}\ -1}$$
$$=\lim_{r\to 0}\frac{r}{\sqrt{r+1}-1}$$
$$=\lim_{r\to 0}\frac{r\left(\sqrt{r+1}+1\right)}{r+1-1}$$
$$=\lim_{r\to 0}\left(\sqrt{r+1}+1\right)=2$$

$\Rightarrow$ f is continuous on $R^2-\{(0,0)\}$.

Example 13：

$$f(x,y)=\begin{cases} \left(x^2+y^2\right)\ln\left(x^2+y^2\right), & (x,y)\neq(0,0). \\ a, & (x,y)=(0,0). \end{cases}$$

Choose an a so that f is continuous at $(0,0)$.

Solution：

$$\lim_{(x,y)\to 0}\left(x^2+y^2\right)\ln\left(x^2+y^2\right)=\lim_{r\to 0^+} r\ln r=0$$
$$\Rightarrow a=0$$